JON MORGAN

Principal – Building Performance & Climate Resilience

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Building physics and simulation engineer with 30 years' experience. Started in aerospace and defence analysis at DSTO and offshore blast modelling at BMT, spent ten years at Arup, including Australasian building physics leadership, and later eight years co-directing an independent ESD practice that delivered rating and compliance submissions for councils and developers. Tests buildings against heat, wind, daylight and energy performance to find where they break down, and builds the tools that make results accurate, consistent and repeatable across every project. Now focused on climate resilience work that asks how buildings will cope with future climate, not only how they perform under today's conditions.

CORE COMPETENCIES

- **Climate Resilience & Adaptation** — heat resilience for buildings and vulnerable populations: heatwave and future-climate stress-testing, heat-resilient public-housing design and retrofit modelling, and technical input on heatwave refuges and building targets for the City of Melbourne.
- **Regulatory & Rating-Framework Delivery** — NCC Part J, Green Star, NABERS, BESS and JV3 compliance and certification delivered commercially at scale.
- **Commercial & Operational Ownership** — co-ran an independent ESD consulting practice for eight years, closely involved in service-line profitability, cost management, strategy, branding and team leadership.
- **Data & Digital Delivery** — built the data and software behind practice delivery, including an automated reporting pipeline across ~10 report types and decision tools used by non-technical staff. Works AI-first: specifies, reviews and tests AI-generated code, and builds LLM pipelines over practice data under controls that keep the output factual.
- **Systems Engineering & Diagnostic Modelling** — aerospace/defence-trained systems engineer; applies formal modelling method (abstraction, parameterisation, verification, validation) across aerospace, offshore, aviation, urban-microclimate and building-physics domains.
- **Carbon & Emissions Profiling** — community and organisational emissions inventories, and upfront embodied-carbon quantification for Green Star submissions; NABERS Upfront Carbon Accredited Assessor.

PROFESSIONAL EXPERIENCE

Visergy — Principal Sep 2026 - present, Melbourne

Relaunched Visergy as a sole trader, offering bespoke façade engineering solutions.

Co-Perform — Principal Jul 2018 - Aug 2026, Melbourne

Co-directed, with a team of four, an independent ESD and building performance consultancy, delivering strategic advisory and technical work for councils, architects, builders and developers. Advisory work included early design advice that let design teams select compliant façades quickly, alongside heatwave and energy-storage advice for councils. Delivery spanned NCC Part J performance solutions (JV3), technical oversight and QA of NatHERS ratings from single dwellings to a 70-storey tower, NABERS Commitment Agreement modelling and Green Star upfront carbon submissions. Co-led the practice's business strategy and rebrand, owned its full analysis stack and built its reporting platform end-to-end.

Moreland Energy Foundation — Senior Consultant Sep 2014 - Jul 2018, Melbourne

Senior Consultant, with one direct report, at a community-focused not-for-profit, delivering energy, emissions and housing work for state government, councils and communities. Built the data platform preparing field teams for the

EnergySmart retrofit program. Produced Hepburn Shire's community emissions baseline, presenting it at a community information session, and annual carbon inventories for a regional council alliance. Engaged directly with households and front-line council staff on solar and energy programs.

Visergy — Principal Sep 2014 - Jul 2018, Melbourne

Operated as an independent sole trader under Visergy, holding every client relationship directly. Delivered JV3, BESS, energy and daylight assessments for town planning and building approvals as a sub-consultant to ESD firms: nine clients over four years, with repeat work from Co-Perform (the largest, which later led to a co-directorship) and EcoResults — concurrent with a Senior Consultant role at Moreland Energy Foundation.

Arup — Australasian Building Physics Lead 2004 - Aug 2014, London / Melbourne

Consulting Engineer in London (2004–05), then Australasian Building Physics Lead in Melbourne (Senior Consulting Engineer grade), with two direct reports and informal mentoring of around ten analysts embedded in mechanical and façade teams across APAC. Owned the simulation infrastructure, software and workflows for energy, daylight, reflected-glare, CFD, plume-dispersion and façade work, including a compute cluster shared with fire engineering that sped up analysis Australasia-wide. Collaborated with structural, mechanical, façade and lighting engineers on commercial, residential, aged-care, master-planning and transport projects, focused on the comfort and safety of occupants, pedestrians and transport users.

BMT Fluid Mechanics — Consulting Engineer 2002 - 2004, Teddington, UK

Consulting Engineer in Teddington, running computer simulation and wind tunnel testing for built-environment and offshore infrastructure. Built the offshore platform blast modelling and CFD pipeline automation behind blast-overpressure studies, and ran physical wind tunnel validation for pedestrian wind microclimate assessment.

Early Career — DSTO, Scott Wilson Irwin Johnston 1996 - 2002, Melbourne

Aerospace and defence engineering at DSTO, then building physics and fluid mechanics consulting at Scott Wilson Irwin Johnston: energy, daylight and ventilation simulation, CFD and wind tunnel testing.

SELECTED PROJECT HIGHLIGHTS

- **Climate-Adapted Public Housing (Victorian DHHS)** — thermal modelling lead in an MEFL-led consortium with DesignInc, where modelling shaped window sizes, shading and envelope performance for heat-resilient public housing for vulnerable tenants; three prototype units are now built and occupied at Mt Martha.
- **Future Climate Weather File Application & Stress Testing** — applied synthetic future-climate weather files to dynamic energy models, quantifying HVAC load and overheating risk in free-running buildings.
- **Public Housing Retrofit Audit Platform (Victorian DHHS)** — built a mobile audit platform combining building condition, tenant experience and billing data with in-home temperature logging to prioritise state-funded retrofits.
- **Arup Building Physics Portfolio (2004–14)** — dozens of daylight and solar-access assessments, pedestrian-wind CFD studies as numerical methods began to replace wind-tunnel testing, an automated replacement for the manual Hassall glare method on Melbourne's rail corridors, and pitch-microclimate studies for about half a dozen major stadia.
- **Compliance & Ratings Portfolio (Co-Perform, 2018–26)** — performance solutions and ratings accepted on 20+ projects. As technical lead on NCC Part J, NABERS Commitment Agreements and Green Star energy and daylight work, and technical oversight of NatHERS ratings; set the modelling method and QA framework that accredited assessors worked within, and held to defensible assumptions when clients pushed for a better result.
- **City of Melbourne Sustainable Buildings & Heatwave Refuges** — technical input adopted into the City's revised sustainable building guidelines for all its internal projects. In collaboration with co-director, provided the heatwave-

refuge advice (smoke ventilation, backup power, solar and battery) and Green Star targets for new council buildings through Nation Partners as lead consultant.

- **Practice Data Model & Reporting Engine** — replaced manual report assembly with a single-source project database (3,000+ fields, entered once) feeding all ~10 client report types, removing copy-paste errors from deliverables.

CREDENTIALS

- NABERS Upfront Carbon Accredited Assessor (2026)
- Advisory Board Member, Energy & Sustainability Hub — Building Designers Association of Victoria (2015–2018)

EDUCATION

- Master of Urban Planning and Environment — RMIT University (2009–2014)
- Bachelor of Engineering (Aerospace) — RMIT University (1991–1995)